Andrei-Cristian Cojocaru

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Education

University of California, Berkeley, Berkeley, CA — Expected May 2027
B.A., Computer Science
Selected coursework: Natural Language Processing, Reinforcement Learning, Deep Learning, Distributed Systems, Information Theory, Convex Optimization

Research Interests

Evaluation and post-training for LLM agents, with a focus on automated graders, adaptive failure taxonomies, gold-free candidate ranking from execution traces, and prompt and program optimization. I study how structured diagnoses of failure can provide feedback for search, reinforcement learning, and continual adaptation.

Publications

Conference papers

[1] W. Ma, **A. Cojocaru**, N. Kolhe, B. Louie, R. S. Sharif, H. Zhang, V. Zhuang, M. Zaharia, and S. Min. "Reliable Fine-Grained Evaluation of Natural Language Math Proofs." International Conference on Learning Representations (ICLR), 2026. arXiv:2510.13888

Workshop papers

[2] M. Cemri*, **A. Cojocaru***, M. Pan, S. Liu, S. Agarwal, A. Krentsel, J. Tang, K. Ramchandran, J. E. Gonzalez, M. Zaharia, A. Dimakis, and I. Stoica. "Fantastic Adaptive Taxonomies and How to Use Them." Workshop on Failure Modes in Agentic AI (FAGEN) at ICML, 2026. [**Best Paper Award**; *equal contribution] arXiv:2607.16387

Research articles

[3] **A. Cojocaru***, L. A. Agrawal*, J. E. Gonzalez, D. Klein, K. Sen, A. G. Dimakis, I. Stoica, and M. Zaharia. "Learned Error Diagnosis for GEPA's Reflection." Research article and reproducibility release, GEPA project, 2026. [*equal contribution] andreiberkeley.github.io/gepa-named-failure-modes

Open-Source Software

AdaMAST. Open-source library for adaptive failure taxonomies; built the generation pipeline, the AdaEvolve and agent-runtime integrations, and the Claude Code and Codex integrations. github.com/multi-agent-systems-failure-taxonomy/AdaMAST

gepa-named-failure-modes. Open-source implementation and reproducibility release for learned error diagnosis in GEPA; an upstream hook is proposed to gepa-ai/gepa (#447, under review). github.com/AndreiBerkeley/gepa-named-failure-modes

Research Experience

BAIR / Sky Computing Lab, UC Berkeley — Undergraduate Researcher — April 2025 – Present

FailureRank — Lead Researcher — August 2026 – Present
Advisor: Professor Ion Stoica
- Lead the design and implementation of a gold-free framework that ranks candidate LLM programs from execution traces and estimates their performance across repeated runs and unseen tasks.
- Test whether structured failure evidence can replace gold-answer scoring for candidate selection in evolutionary optimization and related learning pipelines, and for certification and robustness assessment.
- Develop purpose-conditioned taxonomy generation, in which the evaluation objective (correctness, robustness, feedback quality, or style compliance) determines which failure modes are induced from traces.

Learned Error Diagnosis for GEPA — Equal-Contribution First Author — August 2026 – Present
With the GEPA team; senior author Professor Matei Zaharia
- Designed and built the end-to-end method: an outcome-blind LLM judge that diagnoses failure modes from execution traces, AdaMAST-based taxonomy generation and reduction, and enrichment of GEPA's reflection data. Implemented trace collection, caching, parallel judging, cost accounting, and held-out multi-seed evaluation.
- Under matched $60-per-seed online budgets on HotpotQA, IFBench, and HoVer, failure-mode feedback improved held-out scores by 2.3, 8.6, and 9.7 percentage points and won all nine paired-seed comparisons.
- Designed the *reflective_dataset_enricher* interface and proposed it upstream to GEPA (gepa-ai/gepa #447), allowing existing adapters to use the method without modification.

AdaMAST — Equal-Contribution Co-First Author | November 2025 – Present

Advisor: Professor Ion Stoica

- With Mert Cemri, developed the adaptive-taxonomy direction from his initial taxonomy concept; defined what adaptation means for an evolving agentic system, which failure categories should change, and how to induce them from observed behavior.
- Built the end-to-end adaptive taxonomy-generation pipeline and two of the three principal integrations: evolutionary agent-system optimization (AdaEvolve) and live agent-runtime feedback.
- Originated and implemented the Claude Code and Codex integrations, which expose structured failure evidence during agent execution; contributed extensively to experiments, paper structure, and the post-submission open-source release.

ProofGrader / ProofBench — Researcher and Co-author | April – September 2025

Advisor: Professor Sewon Min; with Professor Matei Zaharia

- Contributed to early ideation with Sewon Min and Wenjie Ma on reliable, fine-grained grading of Olympiad-level natural-language proofs; built the initial marking-scheme generation pipeline.
- Produced core expert annotations and qualitative analysis for ProofBench (435 proofs, 145 problems) and used them to refine the marking-scheme structure and evaluation protocol.
- Developed and ran ProofGrader variants in parallel with Wenjie Ma and contributed to the final ablations; the evaluator reached 0.926 MAE against expert scores and closed 78% of the best-of-16 gap between a binary evaluator and the human oracle.

Honors and Awards

Best Paper Award, Workshop on Failure Modes in Agentic AI (FAGEN) at ICML | 2026

Presentations

- "Reliable Fine-Grained Evaluation of Natural Language Math Proofs." Poster, ICLR 2026, Brazil.

Industry Experience

Mercor — Machine Learning Developer, Remote | May – November 2025

- Built evaluation and data-quality pipelines for language and vision models, combining annotation guidelines, synthetic-data generation, and CI/CD quality gates; reduced annotation errors by 35%.

Logitech — Software Engineering Intern, San Francisco, CA | July – December 2024

- Developed an AI-assisted AWS EC2 optimization system (Cost Explorer, SageMaker, Lambda) that reduced operating costs by 22% and automated 95% of resource-allocation decisions; analyzed ~1 TB of operational data with SQL and Spark.

Outlier AI — Machine Learning Developer, Remote | April – November 2024

- Optimized PyTorch inference with retrieval caching and batch processing, reducing latency by 45% and increasing throughput by 2.8× while preserving output quality.

Makers Fund — Technical Research Project | May – July 2024

- Built a blockchain-gaming market-intelligence pipeline (DappRadar, Artemis, CoinGecko, spaCy, Pandas) that produced daily summaries and investment signals, reducing manual research time by 40%.

Technical Skills

Languages: Python, C++, SQL, TypeScript, Java

ML and LLM systems: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn; LLM evaluation and judging, prompt and program optimization, AI agents and tool use, benchmark development, multi-seed and cost-matched evaluation

Data and cloud: Pandas, Spark, MongoDB; AWS (EC2, SageMaker, Lambda); Docker, Linux, Airflow, CI/CD, Git